

Designing a DevSecOps Toolchain to Support ETL Workflows in Regulated Industries

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Contents

1	Introduction	3
1.1	Context and Motivation	3
1.2	Problem Statement	3
1.3	Research Objectives	4
1.4	Contributions	4
1.5	Thesis Structure	4
2	Literature Review	6
2.1	Background	6
2.2	Defining Key Concepts	7
2.3	RQ1: ETL Pipelines in Regulated Industries	9
2.4	RQ2: DevSecOps and Governance in the SDLC	9
2.5	RQ3: Comparing the Data Lifecycle and the SDLC	10
2.6	RQ4: Tooling for Auditability and Governance	11
3	Methodology	14
3.1	Research Approach	14
3.2	Design Science Process	14
3.3	Case Study Context	14
3.4	Artifact Design Process	14
3.5	Evaluation Plan	14
3.6	Limitations	15
4	Baseline Pipeline	16
4.1	Overview	16
4.2	Architecture	16
4.3	Functionality	16
4.4	Observed Gaps	16
4.5	Summary	16
5	Proposed Architecture	17
5.1	Design Principles	17
5.2	High-Level Overview	17
5.3	Core Components	17
5.4	Integration with DevSecOps Practices	17
5.5	Expected Outcomes	17
6	Prototype Implementation	18
7	Evaluation	19
8	Discussion	20
9	Conclusion	21

Chapter 1

Introduction

1.1 Context and Motivation

In recent years, the value of data has surpassed that of the software systems built to process it [1, 2]. Organisations increasingly treat data as a product, complete with ownership structures, stewardship responsibilities, and lifecycle management frameworks [1]. This shift has coincided with growing regulatory scrutiny across jurisdictions [1]. Frameworks such as APRA’s Prudential Standards [3–6], alongside global instruments such as the EU’s GDPR and the United States’ HIPAA, demand not only the secure handling of information assets but also transparent processes that demonstrate accountability and control [2].

Extract–Transform–Load (ETL) workflows lie at the heart of this transformation, serving as the operational backbone of analytical systems [7, 8]. As Mahmud and Ikbāl [9] note, modern ETL pipelines have evolved into socio-technical infrastructures that sustain the scalability, reliability, and legitimacy of data-driven decision-making. Yet despite the rise of DataOps, many pipelines remain opaque, lacking end-to-end lineage, reproducibility, and verifiable controls [2]. In regulated environments, such opacity undermines compliance and erodes institutional trust.

Software engineering has already faced similar challenges. DevSecOps — the integration of development, security, and operations — has matured into a discipline embedding verification, automation, and governance across the software delivery lifecycle [10–13]. If data is now the product, DevSecOps offers an architectural and cultural foundation for treating data pipelines with the same rigour once reserved for software systems [14]. This research explores that proposition: how DevSecOps principles can be adapted to strengthen auditability and data governance within ETL workflows.

1.2 Problem Statement

While the DataOps movement has improved automation and observability, its primary focus remains operational efficiency rather than regulatory assurance [9, 15]. Existing orchestration frameworks, though powerful, rarely produce immutable audit trails, reproducible execution states, or cryptographically verifiable lineage. Auditability thus emerges as an accidental property rather than a designed one [16, 17]. As a result, organisations face a persistent gap between the technical execution of pipelines and the evidentiary requirements of regulators [3, 4].

This thesis addresses that gap. It investigates how a *DevSecOps-enabled toolchain* can enhance auditability and data governance across the ETL lifecycle. DevSecOps has already institutionalised

security and compliance for software artifacts, but an equivalent framework for data artifacts has yet to emerge. This absence exposes regulated organisations to compliance risk, weakens governance, and impairs their ability to demonstrate trustworthy data management.

1.3 Research Objectives

This research adopts a Design Science methodology [18] to design, implement, and evaluate a DevSecOps-based toolchain that enhances auditability and governance in ETL workflows. The study follows an iterative cycle of artifact design, prototype construction, and evaluation against explicit metrics for lineage completeness, reproducibility, and tamper detection.

Primary Research Question

How can a DevSecOps toolchain be designed to support ETL workflows in regulated industries, enhancing auditability and data governance?

In the process of answering the above question, this paper explores four sub-questions:

Sub-Questions

1. How do ETL workflows manifest in regulated industries, and what auditability and data governance challenges do they face?
2. How do DevSecOps practices enhance auditability and data governance within the software development lifecycle?
3. In what ways do data lifecycle management practices extend or reshape the software development lifecycle when DevSecOps principles are applied to ETL workflows in regulated environments?
4. What categories of tooling exist in DevSecOps and ETL pipelines, and how do their features support auditability and data governance?

1.4 Contributions

This thesis makes three primary contributions. First, it proposes an **architectural model** for a DevSecOps-enabled ETL toolchain, aligning the data lifecycle with secure software delivery practices to improve auditability and governance. Second, it presents a **prototype implementation** that integrates infrastructure as code, automated security scanning, and immutable logging into an operational ETL environment built with Airflow, Spark, and AWS components. Third, it defines an **evaluation framework** for quantifying auditability and governance through traceability, reproducibility, and tamper-resistance metrics. Together, these contributions demonstrate how DevSecOps principles can be embedded in data-centric systems to enhance compliance and trust.

1.5 Thesis Structure

The remainder of this thesis is organised as follows:

- **Chapter 2** reviews literature on ETL pipelines, data governance, and DevSecOps practices.
- **Chapter 3** explains the Design Science methodology, design constraints, and evaluation criteria.
- **Chapter 4** describes the baseline ETL pipeline and identifies auditability and governance gaps.
- **Chapter 5** presents the proposed DevSecOps-integrated architecture.
- **Chapter 6** outlines the prototype implementation and toolchain configuration.
- **Chapter 7** evaluates the system against defined auditability and governance metrics from Chapter 3.
- **Chapter 8** discusses implications, limitations, and avenues for future research.
- **Chapter 9** concludes with reflections on how DevSecOps can enable trustworthy data governance in regulated environments.

Chapter 2

Literature Review

This chapter reviews the body of knowledge relevant to the design of ETL pipelines in regulated industries and the application of DevSecOps principles. It begins by outlining the context of ETL and governance, defines the key concepts central to this research, and then examines the literature associated with each research question.

2.1 Background

Data pipelines underpin modern analytical systems, forming the connective infrastructure between data sources and analytic platforms [14]. They transform raw data into governed, traceable, and auditable outputs used in applications such as fraud detection, market analysis, compliance reporting, and metadata processing.

Although the extract–transform–load (ETL) process appears simple in theory, it is in practice complex, resource-intensive, and often the most time-consuming element of data warehousing [19]. ETL pipelines act as the “eyes of the system,” feeding reliable data downstream; when lineage is lost or visibility fails, the reliability of the entire process is compromised [8]. Vayyala [8] emphasises that maintaining end-to-end lineage across transformations is vital for sustaining trust in analytics-driven decisions.

In regulated industries, compliance depends as much on how data is processed as on the outcomes produced. Frameworks such as GDPR, SOX, and APRA’s CPS standards require secure, ethical, and well-documented data handling [1, 3–6]. These frameworks place accountability, ownership, and audit trails at the centre of compliance culture.

Auditability, in this context, is the ability to demonstrate compliance through reproducibility, lineage, and tamper-resistant logs [2]. Trustworthy analytics rely not only on integrity of data but also on immutable evidence of correct process execution. Governance provides the organisational scaffolding for these assurances—linking policy to technical enforcement through defined roles, procedures, and controls [1, 2]. Failures in either dimension expose organisations to legal, financial, and reputational risk [1].

DevOps emerged to accelerate software delivery through automation, continuous integration and deployment (CI/CD), and rapid feedback cycles [13]. DevSecOps extends this model by embedding security and compliance across the software development lifecycle. Practices such as automated testing, compliance-as-code, artifact signing, and immutable logging produce verifiable artifacts that can be inspected by regulators [10, 11, 13]. This convergence bridges engineering and governance concerns, establishing compliance as a built-in property of the delivery process.

As data itself becomes a primary strategic asset, surpassing the value of the systems that process it [1, 2], the mechanisms of DevSecOps must be extended to the data pipelines that sustain organisational decision-making.

2.2 Defining Key Concepts

The following subsections define the key concepts underpinning this research: DevSecOps, ETL, data governance, and auditability. Each definition draws from existing literature and sets the theoretical foundation for later analysis.

DevSecOps

DevSecOps is the proactive and continuous enforcement of security and compliance requirements throughout the software development lifecycle (SDLC) [10]. Unlike traditional models that treat security as an external stage or afterthought, DevSecOps embeds it within every phase of the lifecycle [12]. Compliance becomes the default state, and deviations from defined standards must be explicitly justified [11].

A defining characteristic of DevSecOps is its reliance on automation to enforce non-functional requirements such as auditability and governance. Through automated testing, provisioning, monitoring, and policy enforcement, systems produce immutable evidence—signed artifacts, logs, and metrics—that regulators and auditors can independently assess [10, 20]. This automated assurance bridges the concerns of development teams and governance stakeholders [12].

By combining cultural expectations of shared responsibility with technical mechanisms for verification, DevSecOps establishes a delivery model in which security and compliance are continuous, measurable, and intrinsic—rather than externally imposed.

ETL

Extract–Transform–Load (ETL) is a structured process that ingests, transforms, and persists data into target repositories such as data warehouses, lakes, or file stores [19]. It traditionally comprises three stages: extraction, where data is sourced from heterogeneous systems; transformation, where it is cleansed, standardised, integrated, and reshaped into coherent semantic structures; and loading, where transformed data is stored for downstream analysis [19]. Kimball and Caserta describe ETL as both the most resource-intensive and most critical link between operational data and analytical use.

Modern practice extends beyond this classical batch model. Approaches such as extract–load–transform (ELT) and hybrid streaming–batch architectures leverage distributed and cloud-native infrastructure [9]. ETL now encompasses not only data integration and quality but also governance and compliance. In regulated industries, pipelines must guarantee lineage, reproducibility, and auditability in addition to performance and scalability [9].

ETL is thus best understood as a flexible socio-technical process. It concludes when data is made persistent and reusable but intersects with broader lifecycle concerns such as orchestration, monitoring, and policy enforcement. In this sense, ETL pipelines underpin both analytical scalability and institutional trust [9].

Data Governance

Data governance provides the strategic framework that defines how data is collected, managed, and protected across an organisation. It establishes the policies, roles, standards, and processes that ensure information assets remain accurate, consistent, secure, and compliant with internal and external regulations [1, 2]. Nai, Rifai, and Sadiq [1] describe governance as a structured framework that ensures the availability, integrity, usability, and security of data through stewardship and accountability roles.

Vayyala [2] expands this understanding by positioning governance as the foundation of reliable data ecosystems—linking policy to technical enforcement through automation, lineage tracking, and immutable audit logs. In practice, governance institutionalises compliance and accountability, transforming data into a verifiable strategic asset that can withstand regulatory scrutiny.

Auditability

Auditability is the designed capability of a system to allow its processes, controls, and outcomes to be observed and verified by stakeholders. Weigand and Elsas [17] define auditability through five interdependent requirements:

- **Normativity:** a clear regulatory or contractual framework that governs operations.
- **Accountability:** verifiable statements linking actions to responsible agents and governing norms.
- **Referentiality Infrastructure:** records that connect operations to their original points of recording.
- **Reliability:** faithful and tamper-resistant representation of events.
- **Verifiability:** the ability of external parties to independently test or validate claims.

Although these principles originate in organisational and financial control, they can be transferred to information systems. Extending this theoretical foundation, Wang et al. [16] demonstrate how auditability can be operationalised through third-party verification, cryptographic proofs of integrity, and support for dynamic data operations. Their work illustrates that auditability is not merely an organisational goal but a property that can be engineered into data management infrastructures.

Applied to ETL pipelines and DevSecOps practices, these five dimensions map directly to regulated-industry concerns:

- **Normativity:** embedding compliance and data-protection policies within workflow execution.
- **Accountability:** linking each transformation or deployment to authenticated agents or automated processes.
- **Referentiality Infrastructure:** ensuring full lineage tracking across extraction, transformation, and loading.
- **Reliability:** maintaining immutable logs, signed artifacts, and cryptographic hashes.
- **Verifiability:** enabling reproducibility tests and independent audits via CI/CD-integrated checks.

Together, these adapted requirements frame auditability as a designed and engineered property of ETL systems. Every data operation and system change must be traceable, reproducible, and independently verifiable against regulatory expectations.

2.3 RQ1: ETL Pipelines in Regulated Industries

ETL pipelines in regulated industries operate within a constant tension between agility and accountability. They are expected to deliver analytical value with speed and scale, yet must simultaneously produce verifiable evidence that every data movement complies with institutional and regulatory constraints. The modern pipeline has therefore evolved into a socio-technical infrastructure—one that mediates trust between engineers, auditors, and regulators [8, 9]. Its success cannot be judged solely by throughput or latency; rather, by its ability to maintain lineage, transparency, and compliance under continuous change.

Financial, healthcare, and critical-infrastructure systems exemplify this dual demand. Frameworks such as APRA’s CPS 230, CPS 234, and CPS 510 [3–5] institutionalise separation of duties and controlled promotion of change, translating ethical responsibility into technical procedure. These standards do not exist apart from engineering practice—they shape it—embedding legal accountability into operational design. Governance scholarship reaches similar conclusions, emphasising that immutable logging, role-based accountability, and retention aligned with compliance lifecycles form the practical machinery of institutional trust [1, 2, 7]. Cloud-native orchestrators such as Airflow, AWS Glue, and Azure Data Factory provide elasticity and automation, but they operate under the shadow of these mandates. Their convenience is constrained by the need to preserve traceability and access control consistent with CPG 235 [6]. In this context, orchestration itself becomes a form of governance: a choreography of tasks and dependencies that leaves a verifiable trail of execution [21].

Over time, this operational discipline has turned the pipeline from a means of data transport into a living archive of compliance. Continuous validation, versioning, and policy enforcement integrate audit logic directly into the runtime environment [2]. Metadata management forms the connective tissue between policy and process, ensuring that provenance, schema evolution, and transformation logic can be reconstructed and defended [7]. Yet even within mature systems, full auditability remains elusive. Toolchain heterogeneity fragments lineage across layers of orchestration, computation, and storage. Business logic embedded in SQL or user-defined functions escapes formal tracing, while schema drift and inconsistent metadata propagation corrode reliability [9]. Few frameworks create immutable, cryptographically-verifiable artifacts by default [7, 17]. What is missing is not information, but integrity—the ability to prove that recorded operations are both complete and untampered. The fragility of this evidentiary chain provides the practical motivation for integrating DevSecOps assurance into ETL environments, turning verification from an external audit to an intrinsic system property.

2.4 RQ2: DevSecOps and Governance in the SDLC

The software development lifecycle (SDLC) treats change as a controlled and observable process. Standards such as ISO/IEC/IEEE 12207 and 15288 define the artifacts, procedures, and documentation that make each phase accountable [22, 23]. Governance in this model is not an external overlay—it is embedded in the way systems are planned, built, and released.

DevSecOps strengthens this principle by embedding assurance, security, and compliance across every phase of delivery [10, 24]. The delivery pipeline becomes a generator of evidence: each build, test, and deployment produces signed artifacts, immutable logs, and compliance manifests that bind design intent to execution [10, 12, 17, 25]. These artifacts collectively form a continuous chain of evidence that makes system state traceable and verifiable at any point in time.

Automation enables this continuity. Infrastructure-as-code (IaC), continuous-integration/continuous-delivery (CI/CD), and policy-as-code (PaC) apply verification directly at the point of change. Static and dynamic testing, dependency scanning, and approval gates transform compliance into an ongoing signal rather than a periodic audit [12, 23, 26]. The NIST Secure Software Development Framework

codifies this expectation: traceable metadata, provenance, and policy enforcement are standard lifecycle outputs [24]. Mature build systems act as provenance engines, linking requirements, commits, and runtime behaviour within the same evidentiary record.

As software supply chains expand, continuous verification of component origin and integrity becomes essential [24, 26]. Toolchains respond through automated scanning, provenance analysis, and AI-assisted policy enforcement [27, 28]. Requirement traceability then links regulatory expectations directly to runtime artifacts [22, 25], ensuring that what is built, deployed, and audited aligns to the same authoritative record [13].

GitOps extends these principles into operations. Desired system state is declared in version control; automated controllers reconcile live infrastructure to that state. Each operational change flows through a commit, retaining authorship, justification, and policy validation [24, 29]. Drift, reconciliation, and approval create a unified provenance trail across both build and runtime [10, 25]. By codifying infrastructure and operations together, GitOps turns system configuration into auditable evidence. The same commit that defines infrastructure also defines the boundary of permitted change. This symmetry between code provenance and data lineage leads directly into RQ3 [30, 31].

Together, DevSecOps and GitOps evolve the SDLC into a process of continuous assurance. Compliance is not declared—it is produced automatically by the system itself.

2.5 RQ3: Comparing the Data Lifecycle and the SDLC

The data lifecycle (DLC) aims for control and reliability but lacks the SDLC’s formal standardisation. Wing describes the DLC as a dynamic, context-dependent model rather than a fixed sequence of phases [30]. Most frameworks share the same goal: to keep data reliable, accessible, and verifiable throughout its lifespan [32, 33]. The SDLC governs change and transformation; the DLC governs persistence and stewardship. One manages evolution, the other preserves continuity.

Despite their differences, the two lifecycles can be aligned. DAMA-DMBOK organises data governance around roles, metadata catalogues, lineage records, and quality assessments that together form an auditable chain of evidence [34]. These artifacts mirror ISO/IEC/IEEE 15289’s information items [22], showing that the logic of software assurance maps closely to data management. DevSecOps provides the mechanisms for this connection. Continuous validation and automated security controls apply equally to code and data [10, 24]. Lineage manifests, schema definitions, and transformation scripts can therefore be versioned and audited alongside software components.

Encryption, secrets management, and role-based access add cryptographic evidence to this framework [14, 35]. Agile governance pushes data validation earlier in the lifecycle [7, 8, 36], mirroring the shift in DevSecOps toward early verification. These expectations reflect APRA’s prudential standards [3–6], where compliance is evidenced through the operational trace rather than through static reports.

When the SDLC and DLC intersect, they create a shared evidentiary framework. ISO/IEC/IEEE 15289’s artifacts—plans, specifications, procedures, and reports—extend to include lineage documentation, schema evolution, and validation outputs. IaC and policy-as-code make these records executable [12, 13, 37]. Reproducibility, tamper detection, and governance then operate through the same automated pipelines that deliver software.

In regulated ETL workflows, auditability becomes a first-class design property. Each transformation of code or data leaves a signed, immutable record that shows what occurred, who performed it, and why it was authorised.

2.6 RQ4: Tooling for Auditability and Governance

Auditability depends on implementation. The ability to trace, verify, and reproduce each operational event relies on tools that enforce control, capture lineage, and preserve evidence of change. In both DevSecOps and ETL environments, tooling translates regulatory requirements such as those in CPG 235 and CPS 234/510 into machine-verifiable artifacts [3, 5, 6].

This section examines the tool categories that make such evidence possible. In DevSecOps, frameworks for infrastructure-as-code, policy validation, and continuous integration generate signed artifacts and provenance records that prove system integrity. In ETL pipelines, orchestration, metadata management, and validation platforms create traceable datasets and transformation histories that support data governance. Each tool category contributes to the same chain of evidence linking software and data processes into a single, verifiable system of record.

DevSecOps tooling: from code to operations

Infrastructure- and Policy-as-Code

IaC and PaC convert infrastructure and policy into versioned artifacts so that compliance checks run before change. Terraform, CloudFormation, OPA, Checkov, and Conftest generate explicit evidence of desired state and policy evaluation [10, 35]. Key contrasts appear in Table 2.1.

Table 2.1: IaC/PaC tools and the evidence they produce

Tool	Strengths	Limitations
Terraform	Versioned infra; plan/apply logs; OPA integration	Limited native provenance
CloudFormation	CloudTrail linkage; tight IAM controls	Vendor lock-in
OPA / Conftest	Declarative policy; decision logs as evidence	Rule complexity
Checkov	IaC scanning; SBOM export	Limited contextual correlation

CI/CD orchestration

GitHub Actions, GitLab CI, Jenkins, and Azure Pipelines attach identity, review, and signatures to every run; pipeline logs form part of the audit trail [24, 25]. Table 2.2 summarises key trade-offs.

Table 2.2: CI/CD systems as provenance engines

Tool	Strengths	Limitations
GitHub Actions	Signed runs and artifacts; Git provenance	Coarse access audit
GitLab CI	Built-in approvals; compliance reports	Higher ops cost
Jenkins	Extensible audit plugins	Plugin sprawl
Azure Pipelines	RBAC tie-in; policy checks	Cloud specific

SAST, DAST, and SCA

Security scanners create structured evidence of vulnerabilities and dependency lineage. They integrate with CI/CD pipelines and feed policy gates [38, 39]. Table 2.3 summarises trade-offs.

Table 2.3: Security scanning tools and audit artifacts

Tool	Strengths	Limitations
SonarQube	Persistent quality/issue history	Needs tuning; limited container context
OWASP ZAP	Reproducible web tests; open formats	Lower throughput at scale
Snyk	Dependency risk; SBOM generation	Proprietary backend; partial offline
Trivy	Unified SCA/SAST; SBOM export	Narrow policy correlation scope

Testing and validation

Test frameworks provide repeatable proof of expected behaviour; their outputs become verifiable artifacts [24, 39]. Table 2.4 outlines key examples.

Table 2.4: Testing frameworks as reproducible evidence

Framework	Strengths	Limitations
Postman / Newman	Versioned API suites; runnable logs	Manual upkeep of collections
PyTest	Parametrised fixtures; stable outputs	Python-only scope
JUnit	Mature CI integration; clear reports	No native audit metadata

Containers and runtime

Images, admission controls, and runtime events contribute directly to system integrity evidence [10, 13]. Table 2.5 lists key tools and their audit characteristics.

Table 2.5: Container and runtime controls as integrity evidence

Tool	Strengths	Limitations
Docker	Image digests and signatures	Limited fine-grained runtime policy
Kubernetes	Declarative policy; drift detection	Multi-cluster audit complexity
Falco	Real-time event evidence	Rule maintenance overhead
Prisma Cloud / Sysdig	Centralised compliance view	Proprietary; cost

Monitoring and observability

Metrics and logs form the final evidence layer, supporting incident reconstruction and trend analysis [10, 35]. Table 2.6 highlights typical stacks.

Convergence

The convergence of DevSecOps and data engineering tooling turns compliance into a continuous process. GitOps operationalises this model: desired system state is declared in version control, controllers reconcile it automatically, and each reconciliation emits evidence of conformity. Tools such as ArgoCD, Flux, and native cloud operators maintain a single, verifiable source of truth where software, infrastructure, and data share the same provenance chain [10, 24, 29].

In this unified model, the same controls that attest software integrity also attest data integrity. Approval gates in CI/CD govern the promotion of ETL workflows; policy checks constrain data transfers; and container attestations record the runtime context of both applications and transformations.

Table 2.6: Observability stacks and audit support

Tool	Strengths	Limitations
Prometheus	Transparent, timestamped metrics	High cost for long retention
Grafana	Clear dashboards; evidence views	Dependent on source data quality
ELK / Splunk	Searchable forensic logs	Operational cost; index complexity

Observability systems link these artifacts into a continuous audit timeline spanning code, infrastructure, and data. Regulatory expectations under CPG 235 and CPS 234/510 [3, 5] are satisfied not through separate audits but through a single, declarative lifecycle.

Three assurance patterns make this convergence practical. *Single-path promotion* ensures that every artifact—application, configuration, or dataset—follows the same reviewed path to production. *Policy-guarded reconciliation* binds control objectives to each applied change. *Attested runtime* couples container and orchestration evidence to specific commits and policies, closing the loop between declared intent and observed outcome [10, 24].

GitOps alone cannot guarantee perfect lineage or privacy. Gaps appear when commits diverge from data artifacts or when ad-hoc fixes bypass version control. These are governance, not purely technical, failures. They weaken the chain of evidence but can be reduced through strict “no-change-outside-Git” policies, shared identifiers across CI/CD and catalogues, and log retention aligned with audit horizons [29, 36].

When applied consistently, these practices align the software and data lifecycles. DevSecOps automates controls; ETL workflows supply the lineage. Together they create auditable data products supported by a continuous, machine-verifiable chain of evidence.

Synthesis and Related Works

Research at the intersection of DevSecOps and data governance converges on the same foundation—automation, traceability, and verifiable control—but few works treat these as parts of a single evidence system spanning the full ETL lifecycle.

The IGI Global collection *Data Governance, DevSecOps, and Advancements in Modern Software* [1] provides the strongest conceptual foundation. Its chapters explain how metadata management, lineage, and policy-as-code can bridge security automation and governance accountability. Vayyala [7] presents metadata as a governance mechanism; Kose [36] links agile compliance to DevSecOps; and Soussane [14] applies these principles to real-time analytics. Collectively, they demonstrate how DevSecOps can embed governance within delivery pipelines, though they stop short of implementing immutable or reproducible evidence across the data lifecycle.

Subsequent research in software assurance [10, 24] and lifecycle management [30, 31] reinforces these principles but remains domain-bound—software or data, not both. This thesis advances that junction by proposing a unified DevSecOps-enabled ETL toolchain that integrates assurance and governance into one continuous, verifiable chain of evidence for regulated industries.

Chapter 3

Methodology

3.1 Research Approach

Explain the overall design science methodology used. Justify its selection for solving an engineering problem through artifact design.

3.2 Design Science Process

Describe each stage:

- **Problem identification:** Auditability and governance gaps in ETL.
- **Artifact design:** A DevSecOps toolchain architecture.
- **Demonstration:** Implementing the prototype.
- **Evaluation:** Measuring lineage, reproducibility, and tamper detection.

3.3 Case Study Context

Briefly describe the regulated domain (e.g., financial systems, SEC filings dataset). Justify its representativeness and constraints.

3.4 Artifact Design Process

Detail the iterative development process—tools, versioning, and validation loops.

3.5 Evaluation Plan

Summarize how you'll measure auditability, reproducibility, and tamper resistance.

3.6 Limitations

Acknowledge what is out of scope or constrained (e.g., scalability, dataset variety).

Chapter 4

Baseline Pipeline

4.1 Overview

Describe the initial ETL setup (Airflow + Spark + S3, deployed via Docker). Explain its purpose as a comparison point.

4.2 Architecture

Insert a diagram showing the baseline system structure. Highlight key components and data flow.

4.3 Functionality

Detail how data moves through extraction, transformation, and loading.

4.4 Observed Gaps

Discuss deficiencies in:

- Lineage tracking
- Reproducibility
- Governance and security controls

4.5 Summary

Conclude with why the baseline cannot meet auditability expectations.

Chapter 5

Proposed Architecture

5.1 Design Principles

Outline the guiding principles of the proposed DevSecOps-integrated ETL system: automation, immutability, traceability, reproducibility.

5.2 High-Level Overview

Provide an architectural diagram comparing baseline and proposed design.

5.3 Core Components

Explain each layer:

- **Infrastructure as Code:** Terraform for provisioning.
- **Orchestration:** Airflow for controlled execution.
- **Security:** CI/CD checks, artifact signing, SAST/SCA.
- **Governance:** Immutable audit logs, metadata tracking.

5.4 Integration with DevSecOps Practices

Map the proposed pipeline to the DevSecOps lifecycle (plan → code → build → test → release → monitor).

5.5 Expected Outcomes

Summarize the architectural advantages for auditability and compliance.

Chapter 6

Prototype Implementation

To be completed in Thesis B

Chapter 7

Evaluation

To be completed in Thesis B

Chapter 8

Discussion

To be completed in Thesis B

Chapter 9

Conclusion

To be completed in Thesis B

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